MARINE INTERNAL COMBUSTION ENGINES (MME 467)

DUAL FUEL AND GAS ENGINES

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Characteristics of a Diesel Fuel

- The **most important characteristic** of diesel fuel is the **cetane number**, as this indicates how readily the fuel self-ignites.
- **The cetane number measures the ignition quality of diesel fuels.**
- A scale of 0-100 is constructed by assigning a value of 0 to α-methylnaphthalene ($C_{10}H_7CH_3$), a naphthenic compound with poor self-ignition qualities, and a value of 100 to n-cetane ($C_{16}H_{34}$), a straight-chain alkane with good self-ignition qualities.
- A 65-cetane fuel would have an ignition delay performance equivalent to that of a blend of 65 percent n-cetane and 35 percent α-methylnaphthalene by volume.
- **Viscosity is generally an important parameter for diesel fuel. Fuel that is too highly viscous can cause damage to the fuel pump (e.g. cam and follower wear) due to higher pressure.**
- Lower viscosity formulations flow faster and smoother through the oil pump, requiring less energy to transfer lubrication throughout the engine.
- ✓ They reduce friction between the piston compression rings and the cylinder walls, minimizing metal-to-metal contact and prolonging the life of your engine components
- However, too low viscosity may lead to a lack of lubrication.

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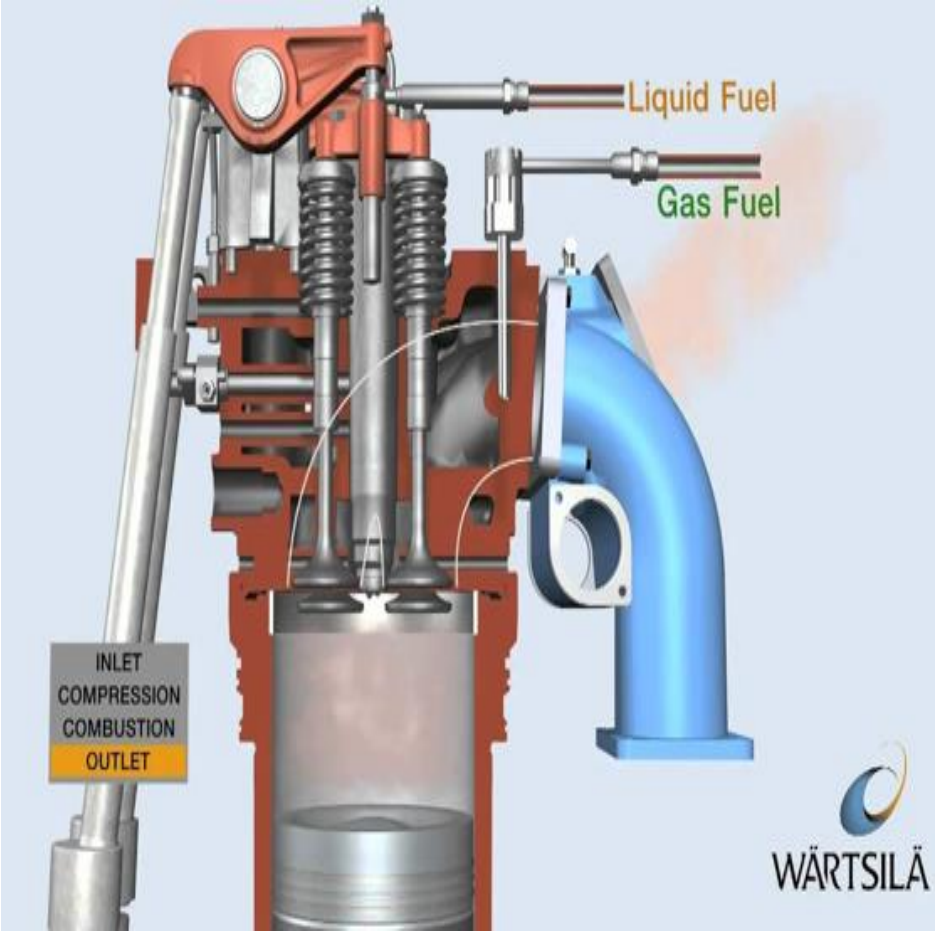
Dual-Fuel (DF) Engines

- Fuel remains one of the highest single cost factors in running a ship and also the source of the most disruptive operating problems.
- For most of the history of the diesel-engined motor ship, there have been no regulations on any aspect of the fuel used beyond a minimum flashpoint of 60°C.
- **The flashpoint is the temperature at which the liquid has to be heated for the vapour to form a combustible mixture with air at atmospheric pressure.**
- The term dual-fuel (DF) engine refers specifically to a diesel engine designed to run on either oil fuels or gas; particularly liquefied natural gas (LNG).
- Operating on natural gas results in very low emissions due to the clean-burning properties of the fuel and its low content of pollutants.
- The main constituent (Methane) is the most efficient hydrocarbon fuel in terms of energy content per amount of carbon, containing the highest amount of hydrogen per unit of energy of all fossil fuels.
- In gas operating mode, the specific carbon dioxide emissions are thus typically reduced by between 15% and 20% compared with heavy fuel oil (HFO)/marine diesel oil operation.

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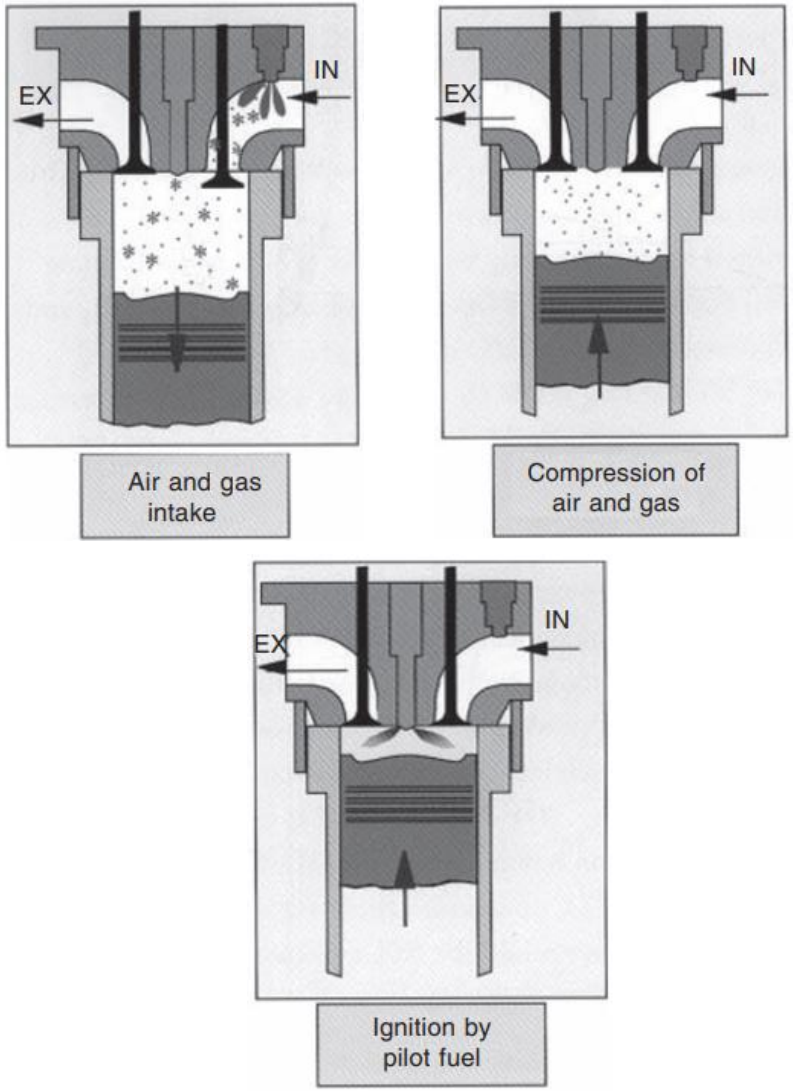
Dual-Fuel (DF) Engines

- The corresponding reduction in nitrogen oxide (NOx) emissions is 85%–90%, while sulphur oxide (SOx) and particulate emissions are almost eliminated.
- Furthermore, there is no visible smoke, no sludge deposits, and no lead emissions; and benzene emissions are reduced by around 97%.
- **Growing opportunities for dual-fuel and gas engines in land and marine power markets have stimulated designs from leading medium-speed and low-speed engine builders.**
- Natural gas has very good combustion characteristics in an engine and as it is lighter than air and has a high ignition temperature, is also considered a very safe fuel.
- **Wartsila Corporation is still considered as the largest manufacturer in the dual-fuel four-stroke sector. Its engines can be run in either gas mode or liquid-fuelled diesel mode.**



Dual-Fuel (DF) Engine (Wartsila)

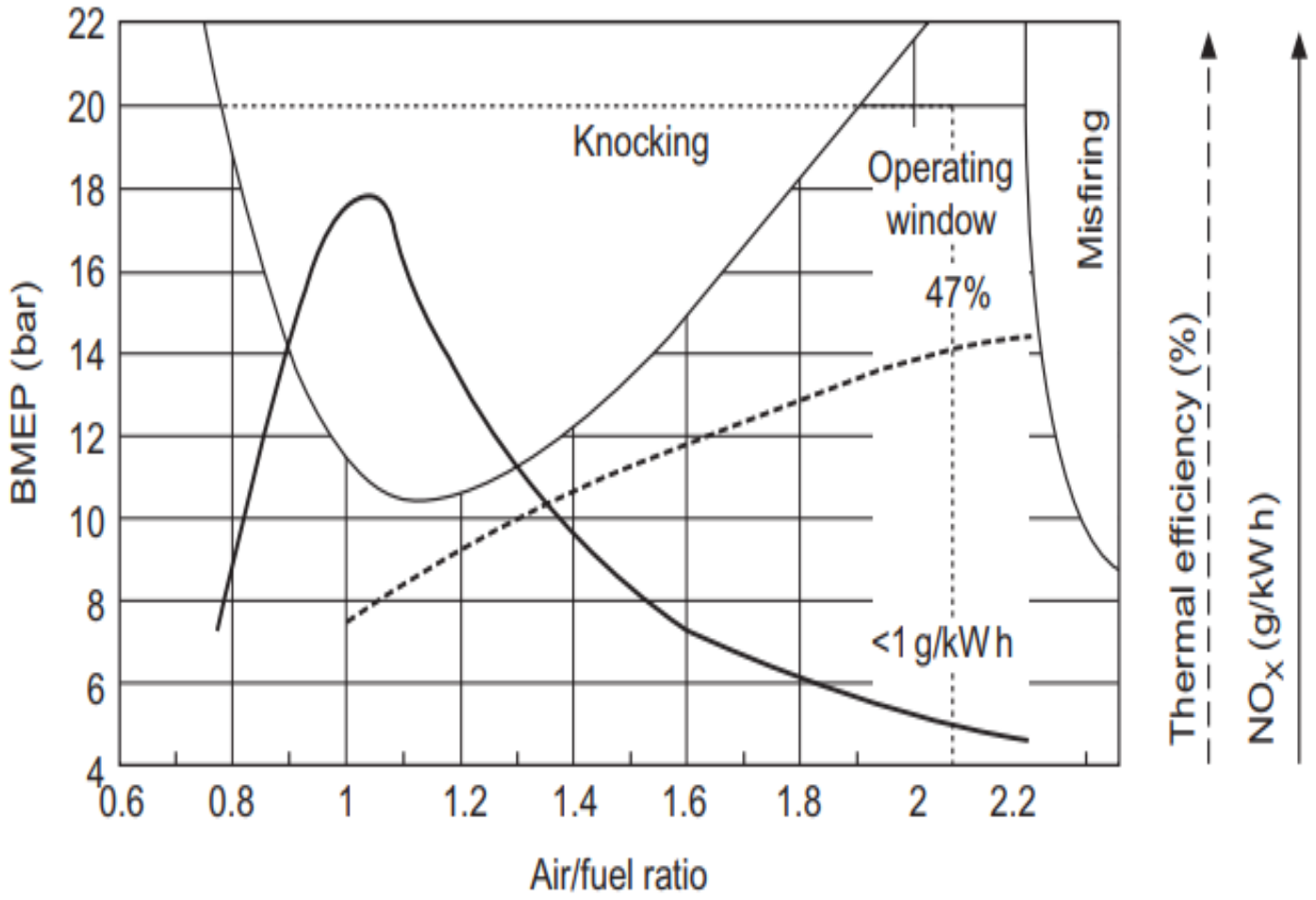
- In **gas mode**, the engines work according to the lean burn principle, with a lean premixed air–gas mixture in the combustion chamber.
- Less NOx is produced and efficiency increases during leaner combustion because of the higher compression ratio and optimized injection timing.
- **A lean mixture is also necessary to avoid knocking (self-ignition).**
- The gas is fed into the cylinder in the air inlet channel during the intake stroke. The lean air–gas mixture is ignited by a small amount of diesel fuel injected into the combustion chamber.
- This high energy source ensures reliable and powerful ignition of the mixture, which is needed when running with a high specific cylinder output and lean air–gas mixture.
- To secure low NOx emissions, it is essential that the amount of injected diesel fuel is very small.



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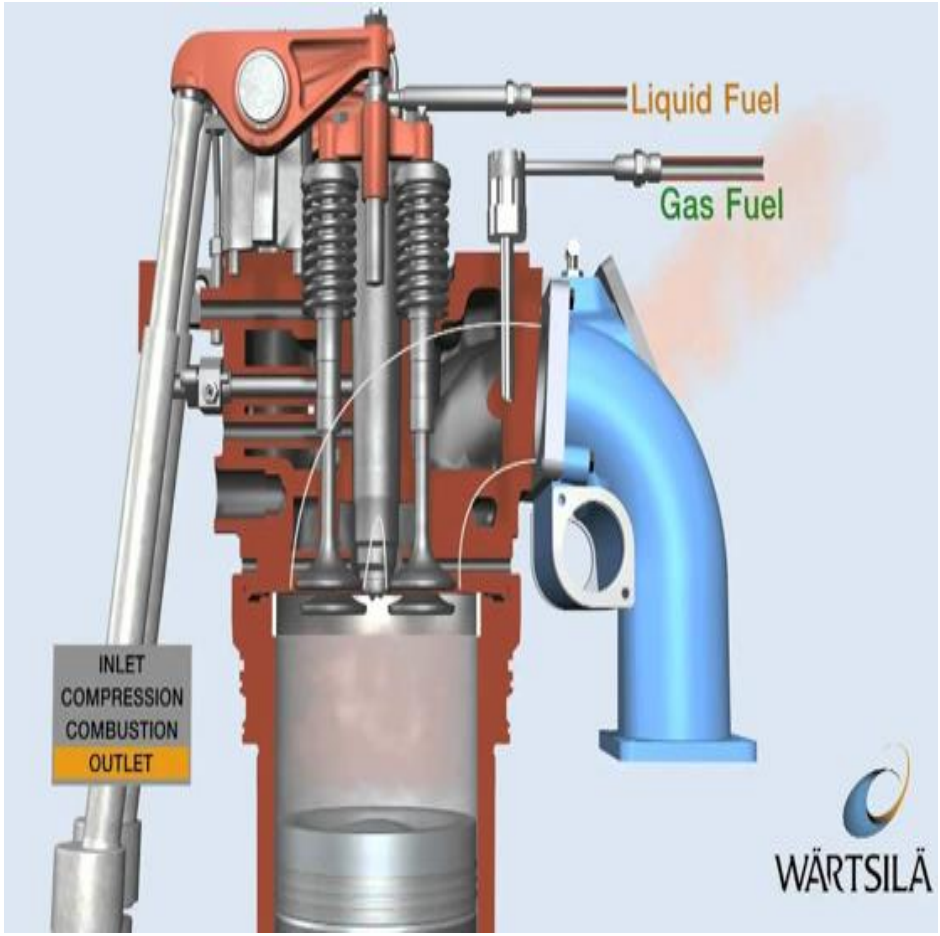
Dual Fuel (DF) Engine (Wartsila)

- When the DF engine is running in gas mode with a premixed air–gas mixture, the combustion must be closely controlled to prevent knocking and misfiring.
- The only reliable way to effect this is to use complete electronic control of both the pilot fuel injection and the gas admission on every cylinder head.
- The global air–fuel ratio is controlled by a wastegate valve, which allows some of the exhaust gases to bypass the turbine of the turbocharger.
- This ensures that the ratio is of the correct value independent of changing ambient conditions such as temperature.



Dual Fuel (DF) Engine (Wartsila)

- In **diesel mode**, the engine works according to the normal diesel concept using a traditional jerk pump fuel injection system: diesel fuel is injected at high pressure into the combustion chamber just before the top dead center (TDC).
- Gas admission is deactivated but the pilot fuel remains activated to ensure reliable pilot ignition when the engine is transferred to gas operation.
- The DF engine is normally started only with the pilot fuel injection enabled to a speed of approximately 60% of the nominal engine speed.
- When the ignition is detected in all cylinders, the gas admission is activated and the engine speed is increased to the nominal speed.
- ✓ This safety measure ensures that no excess unburnt fuel is fed directly into the exhaust gas system.



Wartsila DF Engine Models

- The 32DF and 50DF versions earned contracts for offshore supply vessel and LNG carrier propulsion projects, both burning natural gas.
- The 32DF engine was introduced for marine applications in 2000 to meet the requirements of a new safety class for installations with a gas pressure of less than 10 bar in a single-pipe arrangement.
- The larger 50DF engine was officially launched in the following year, four six-cylinder models (each developing 5700kW) being specified to drive the main gensets of a diesel-electric LNG carrier.
- The 31DF engine was developed taking into account increased interest in new fuel types, reduced lifecycle cost and increased uptime, reduced environmental impact in combination with increased fuel and operational flexibility, and with guaranteed reliability and safe operation.
- With the 31DF, diesel fuel consumption can run as low as 165 g/ kWh, putting it significantly ahead of any other four-stroke diesel engines available in the market.
- The Wartsila 31 can be used both as a main propulsion engine and in diesel-electric configurations, or as an auxiliary engine, and it can be optimized for running either at a constant speed, along a propeller curve, or with constant torque.

Wartsila 50DF Engine

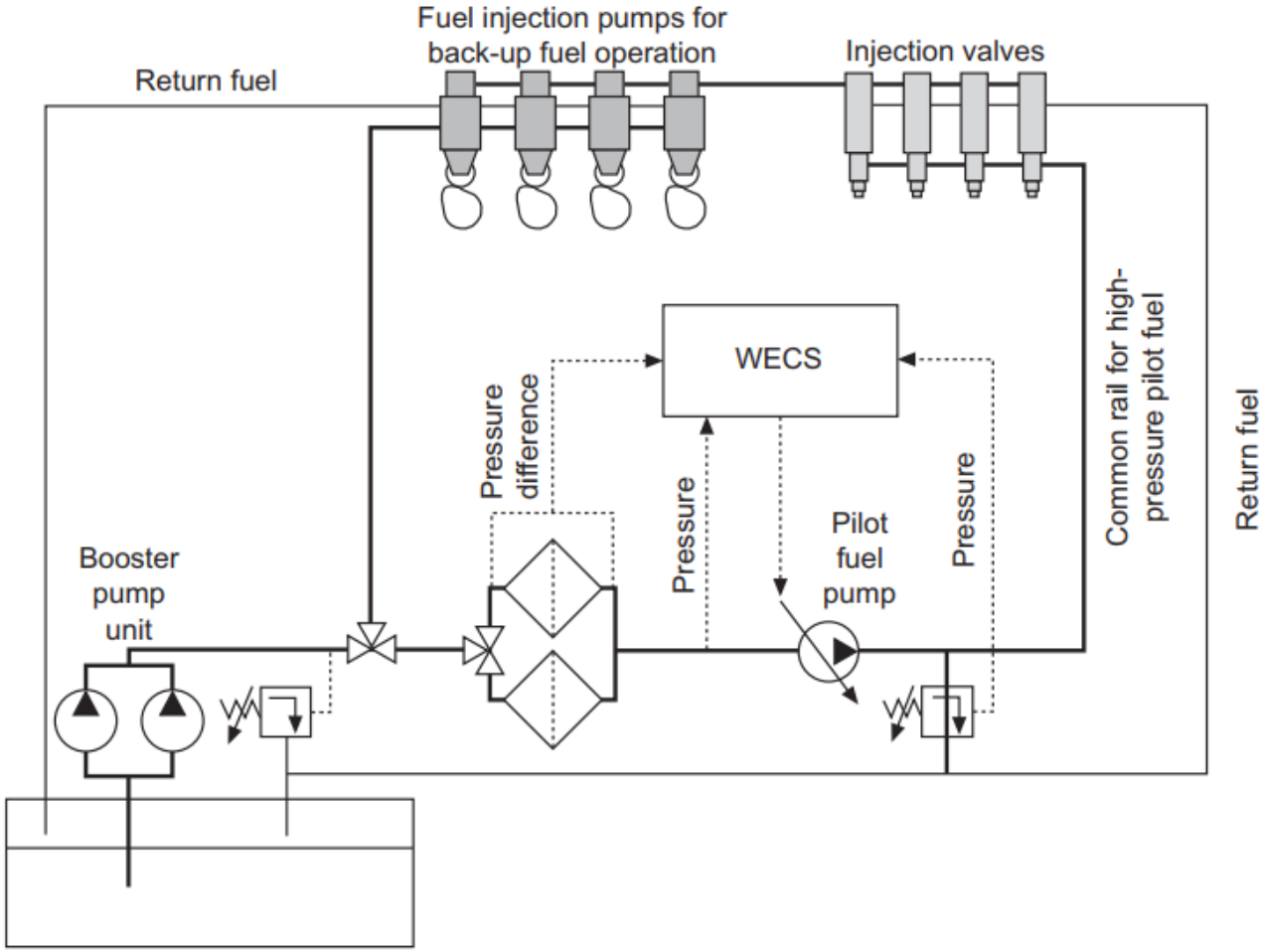
- A thermal efficiency of 47% (higher than any other gas engine at the time of its introduction) is claimed, and the engine delivers the same output whether running on natural gas, light fuel oil, or HFO.
- The air–fuel ratio is very high, typically 2.2:1.
- The mixture is uniform throughout the cylinder as the fuel and air are premixed before introduction into the cylinders, which helps to avoid local NOx formation points within the cylinder.
- Engine starting is always in diesel mode, using both main diesel and pilot fuel. At 300rev/min, the main diesel injection is disabled and the engine is transferred to gas mode.
- Gas admission is activated only when combustion is stable in all cylinders, ensuring safe and reliable starting.
- In gas-running mode, the pilot fuel amounts to, 1% of the full load fuel consumption, and the quantity of pilot fuel is controlled by the Wartsila engine control system (WECS).



50DF engine in V12 cylinder form for LNG carrier propulsion

Wartsila 50DF Engine

- The diesel fuel oil supply on the engine is divided into two systems: one for the pilot fuel and one for the backup fuel.
- The pilot fuel is raised to the required pressure by a pump unit incorporating duplex filters, a pressure regulator, and an engine-driven radial piston-type pump.
- The high-pressure pilot fuel is then distributed through a common rail (CR) pipe to the injection valves at each cylinder; pilot fuel is injected at approx. 900 bar pressure and the timing and duration are electronically controlled.
- Backup fuel is fed to a normal camshaft-driven injection pump by which it is pumped at high pressure to a spring-loaded injection valve of standard design for a diesel engine.



Diesel fuel oil supply system for the Wartsila DF engine.

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Wartsila DF Engine Models

Wärtsilä DF engine data

Model	32DF	34DF	31DF	50DF
Bore	320 mm	340 mm	310 mm	500 mm
Stroke	350 mm	400 mm	430 mm	580 mm
Engine speed (rev/min)	720/750	720/750	720/750	500/514
Mean piston speed (m/s)	8.4/8.75	9.6/10.0	10.32/10.75	9.7/9.9
Mean effective pressure (bar)	19.9	22	27.2	20/19.5
Output/cylinder (kW)	335/350	480/500	530/550	950
Cylinder numbers	6, 9 L 12, 18 V	6, 8, 9 L 12, 16 V	8, 10, 12, 14, 16 V	6, 8, 9 L 12, 16, 18 V
Power band	2010– 6300 kW	2880– 8000 kW	4240– 8800 kW	5700– 17,100 kW

Gas Engines

- ❖ Three main types of gas-burning engines are available to the marine industry which come out of different development directions and hence with some different characteristics.
- ❖ The **spark-ignited "gas only" engines** were first developed for the land-based power industry with simplicity and good overall performance at the lowest total emissions as their prime requirements.
- ✓ These engines are dependent on a permanent gas supply and initially came into the marine industry as engines for short-distance ferries.
- ❖ The **diesel-ignited gas engine** with dual fuel capability was also originally developed for powerplant use where its ability to operate both on liquid and gaseous fuels at high specific power was a particular advantage.
- ✓ The development focus was originally on low NOx emissions at high loads, but recently also on part load performance and variable speed capability.
- ✓ The diesel-ignited gas engine was the first type to establish itself in the marine industry and is currently the dominating engine type in this market.





Gas Engines

- The **diesel gas engine** with direct gas injection first came to use in the offshore industry where its high fuel flexibility and very high power density were of prime attraction.
- This concept is unique in posing no particular requirements to the self-ignition stability of the fuel gas.
- Its diesel operating principle ensures that the combustion of the gas fuel is very complete but at the cost of higher NOx emissions than other gas engine types.
- Gas-burning engines operate according to two different principles, the "pre-mixed" Otto- and direct-injected diesel cycles:





Gas Engines: Otto Gas Engines

- **Spark-ignited gas engines** with either carburettors or port injection of gas. These are single-fuel engines and must therefore meet special redundancy requirements for marine applications.
- **Diesel-ignited gas engines** with conventional low-pressure gas feed but with ignition by the injection of a certain quantity of diesel fuel, also known as the Otto DF or low-pressure DF principle.
- ✓ These will always need a certain quantity of diesel fuel for running even in Gas mode, but on the other hand, they may also run on 100% liquid fuel (diesel- or HFO), i.e. dual fuel capability.

- Otto Gas engines with their homogeneous combustion generally have low NOx emissions and high efficiency.
- However, they require certain stability of the fuel gas against self-ignition and they must be carefully developed in order to keep un-burnt gas to a minimum.
- Spark-ignited and diesel-ignited gas engines show some differences in this respect, especially at part load.

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Gas Engines: Diesel Gas Engines

- The fuel gas is directly injected at high pressure into the cylinder after the diesel pilot fuel has ignited.
- Such engines have dual fuel capability and may also run on 100% liquid fuel (diesel or HFO).
- Diesel gas engines have diffusion burning which ensures the good capability of burning gases with low knocking stability and at the same time producing low emissions.
- However they require a high-pressure gas system (typically 300 bar) and additionally, exhaust after-treatment is needed to comply with IMO Tier III NOx emission limits.

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Gas Engines: Types and Characteristics

- They are small engines that are often derived from automotive gasoline and diesel engines, covering an output range from a few kW up to 500 kW.
- They are generally spark-ignited without a pre-chamber, operating with a stoichiometric mixture and with mixture formation in the intake system.
- Naturally aspirated as well as turbocharged engines are found in this category, some equipped with a three-way catalyst and a sensor for emission control.
- Spark-ignited lean burn engines may either be developed solely for co-generation purposes or derived from small-bore marine diesel engines.
- ✓ Such engines cover an output range from 50-1000 kW up to around 7000 kW.
- ✓ They are often turbocharged lean burn engines with a pre-chamber and with external mixture formation.
- Engine efficiency can be high and NOx emissions so low.
- Gas-injection engines are relatively complicated and need a high-pressure gas compressor.
- In the lower output range, therefore, the costs are hardly justified except where special gases are to be used.
- But for large engines, and for special gases, the high-pressure gas-injection engine is considered the only option.
- Long-term service experience has confirmed that such engines can provide reliable and safe operation.

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